

MASTER'S FINAL PROJECT · DATA SCIENCE · HOLON INSTITUTE OF TECHNOLOGY

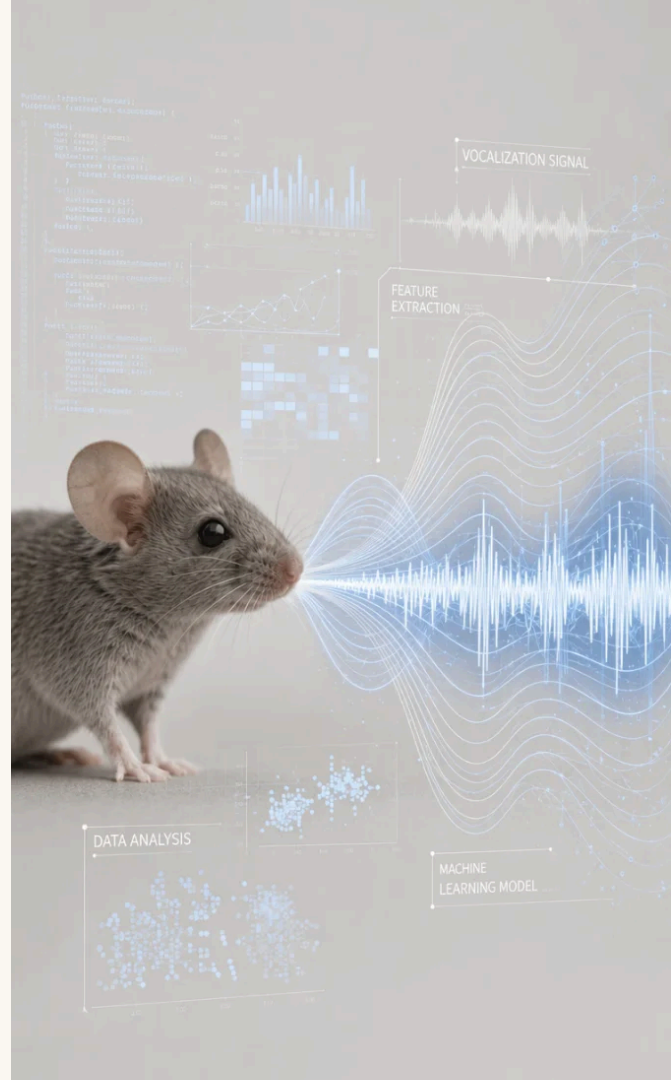
Analysis of Ultrasonic Vocalizations of Mice for Autism Detection

Chen Aharon · Aviel Bitton

Advisor: Dr. Dror Lederman

In collaboration with Prof. Hava M. Golan · Ben-Gurion University

2026



OVERVIEW

Agenda

01 Motivation

Why it matters

02 Background

The problem and the model

03 The solution

USVs as a biomarker

04 Research questions

Aims and hypotheses

05 Dataset

Recordings and cohorts

06 Methods

From audio to typed syllables

07 Models

Tabular & sequence classifiers

08 Results

Performance and baselines

09 Engineering

A reproducible system

10 Conclusions

Answers and next steps

SECTION 01

Motivation – Personal

Seeing a young family member thrive after early, well-matched care — early detection is personal.



SECTION 01

Motivation – Professional

Applying machine learning to real biomedical data — in search of an objective biomarker.



SECTION 02

Background

Autism, the diagnostic gap, and why mouse vocalizations.

02

BACKGROUND

Autism Spectrum Disorder (ASD)

1 in 31

Children diagnosed with autism — CDC (2025)

A complex neurodevelopmental condition, defined by two core behavioral domains:

- Difficulties in social communication
- Restricted, repetitive patterns of behavior

≈4× more common in boys

↑ Diagnoses rising steadily

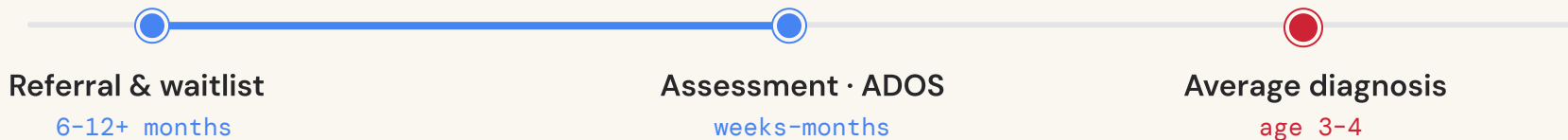


BACKGROUND

The diagnostic gap

Today, autism is diagnosed only through expert behavioral assessment — a slow, multi-step clinical pipeline.

- Subjective, expert-only — no objective test
- Standardized observation such as ADOS, across multiple sessions



BACKGROUND

The need for a biomarker

Diagnosis lands years after infancy,
exactly the window when early intervention matters most.

WHY TODAY'S DIAGNOSIS FALLS SHORT

- Entirely dependent on the clinician's subjective judgment
- Hard to apply reliably in infancy

THE NEED

An objective biomarker

A biological, quantitative measure for early, fast, and independent detection.

Objective

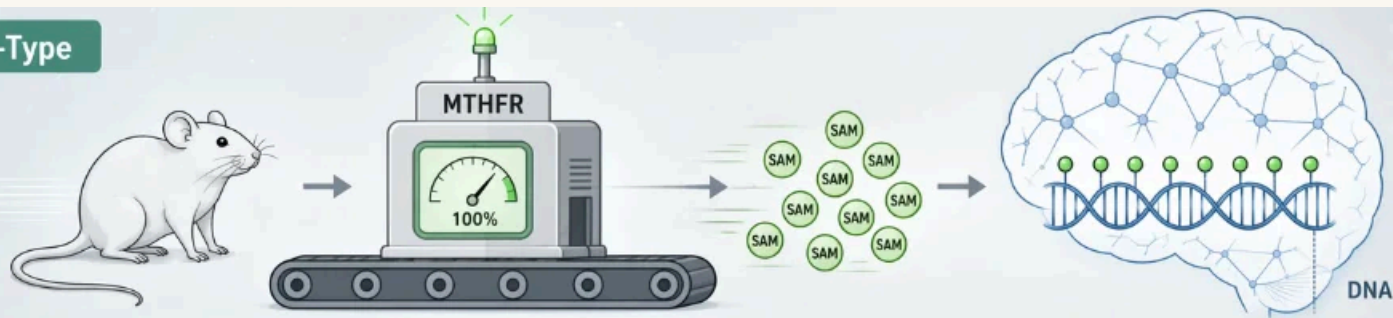
Quantitative

Early

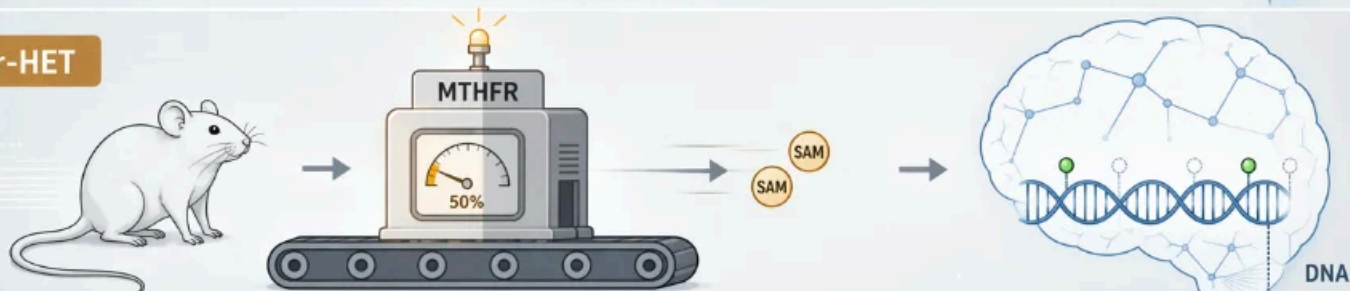
BACKGROUND

How would you model autism in a mouse?

Wild-Type



Mthfr-HET



Not one gene

A recipe of many genes and metabolic steps.

MTHFR mutation

Leaves the enzyme only partially functional.

ASD-like, not autism

Mice show autism-like behavior only.

BACKGROUND

So why do vocal signals matter?

Vocal communication is one of the most basic behavioral outputs in nature, a direct readout of brain activity.



01 Fundamental behavioral output

One of the most basic signals in nature.

02 Speech-like structure

Shares timing and frequency modulation with human speech.

03 Sensitive to neurodevelopment

Brain-development disruptions surface in the calls.

BACKGROUND

Why a mouse?

Four reasons the mouse is the right animal model for a neurodevelopmental disorder.



85–95%

GENETIC HOMOLOGY

Nearly the same genome

Protein-coding genes are shared, and human **MTHFR** has an exact mouse ortholog, **Mthfr** — doing the same biological job.



1 variable

ISOLATED VARIABLES

A clean research instrument

Same genetics, food and housing — the **only** difference between groups is the gene, with none of the environmental noise of humans.



~20 days

FAST LIFECYCLE

Development you can watch

Gestation ~20 days, adulthood in weeks — whole critical windows of brain development observed in a short time, not the 3–4 years a child takes.



USVs

VOCAL COMMUNICATION

A direct link to our signal

Social ultrasonic calls arise from circuits sharing an evolutionary basis with human communication — disrupted calls flag social-communication deficits.

BOTTOM LINE

A mouse is not a small human — but an excellent biological test-bed for how a single gene shapes brain development and social behavior.

SECTION 03

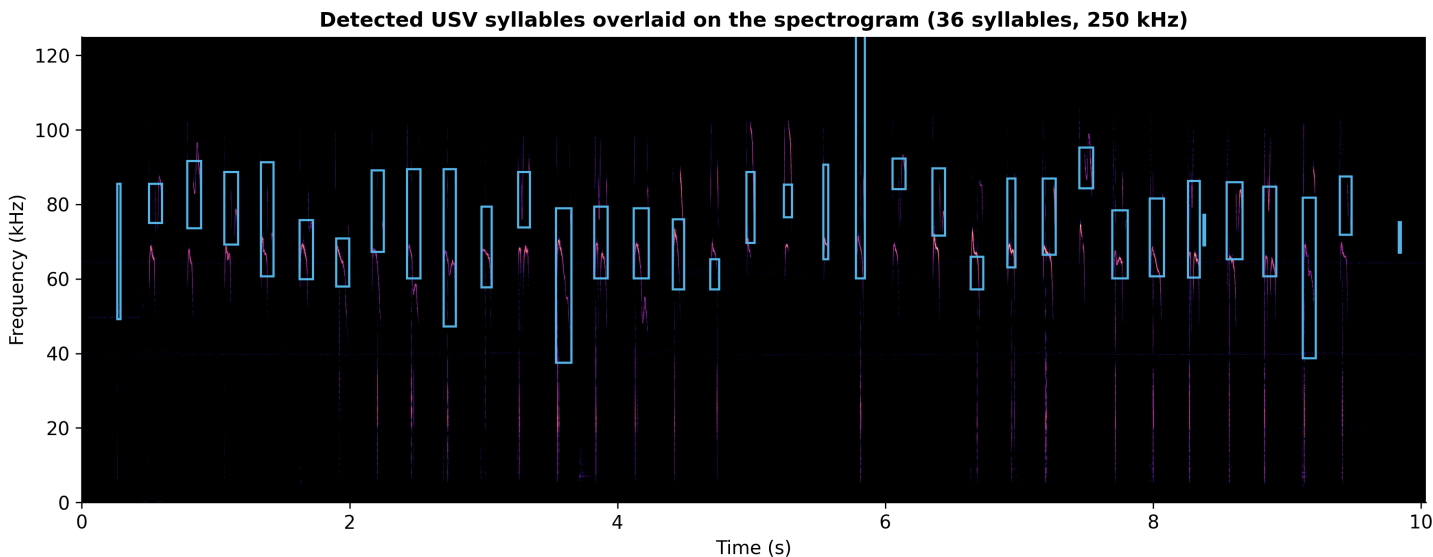
The solution

Mouse ultrasonic vocalizations as an objective biomarker.

03

THE SOLUTION

The candidate biomarker: mouse USVs



35–125 kHz calls

Emitted when pups are isolated.

Earliest signal

The first measurable communicative behavior.

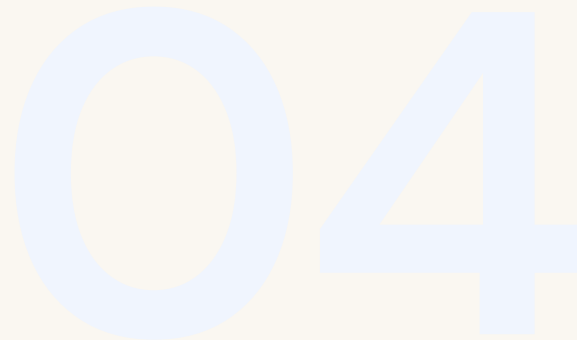
Our hypothesis

A non-invasive, objective, quantifiable ASD biomarker.

SECTION 04

Research questions

Three research questions — models, features,
and generalization.



Three questions frame the study

RQ1

MODELS

Which model family best classifies HET vs wild-type pups from their USVs — XGBoost, TabPFN, or sequence models?

RQ2

FEATURES

Which acoustic characteristics drive the classification — and do two or three consistent features emerge?

RQ3

REGIME & GENERALIZATION

How do the evaluation regime (subject-dependent vs -independent) and cohort (per-strain vs pooled) affect performance — above all, generalization to unseen mice?

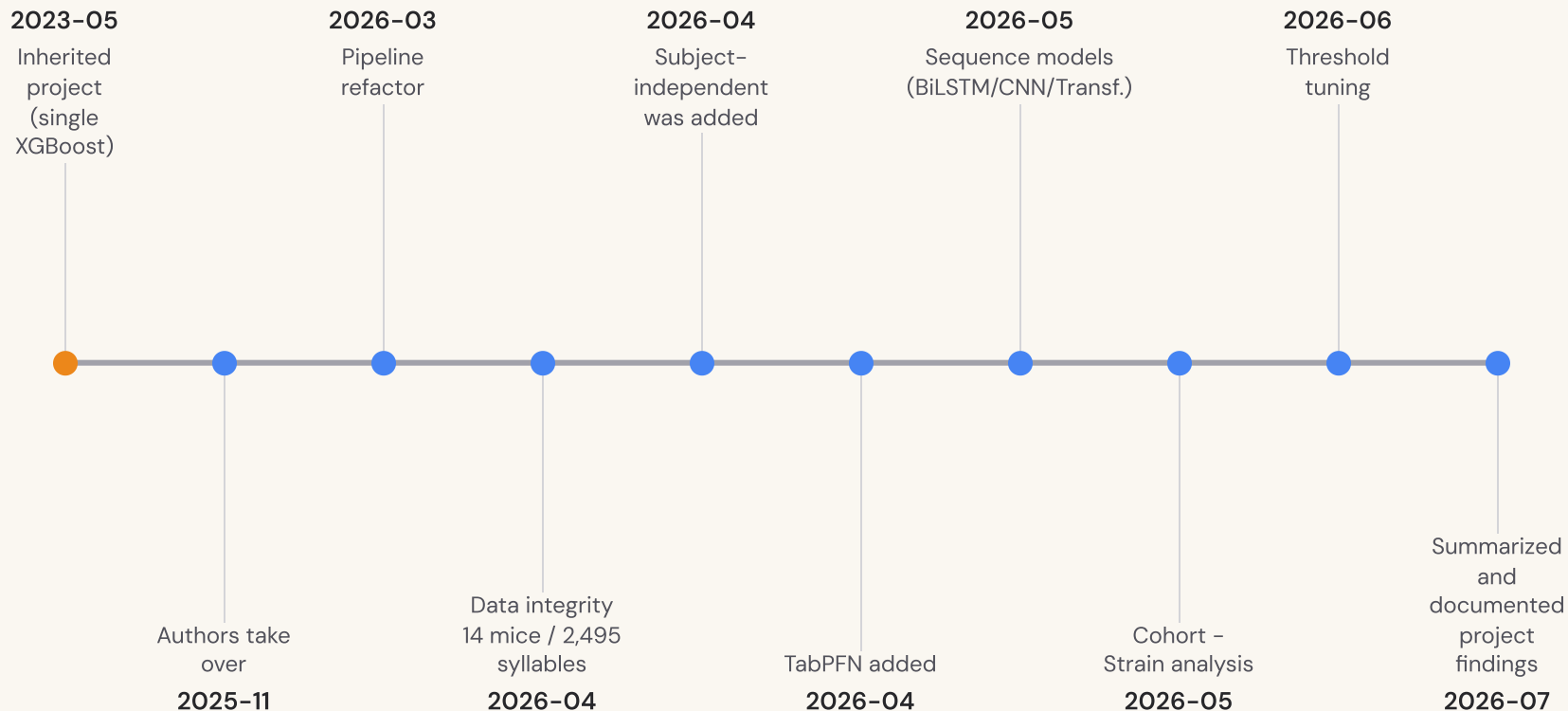


ENGINEERING GOAL

Turn the inherited, non-runnable prototype into a reproducible, documented, end-to-end pipeline built on a single, auditable source of truth.

PROJECT TIMELINE

Reconstructed from the commit history



SECTION 05

Dataset

Recordings, syllables, strains, labels, etc.

05

DATASET

The raw data: metadata & audio

Raw material arrives as thousands of WAV recordings + per-year Excel metadata file.

Mother	Mother genotype	Pup ID	Sex	TARGET Offspring genotype	Day (P4–P12)	Session	Recording no.
O8001P	HT	O8130I	F	HT	P12	1	T0000001 .wav
O8017Q	WT	O8206D	M	WT	P8	2	T0031544 .wav
15044B	HT	15229K	F	HT	P6	1	T0102337 .wav

Highlighted column = the target label · HT (model-autism) vs WT (healthy).

126
pups

408
sessions

125,576
syllables

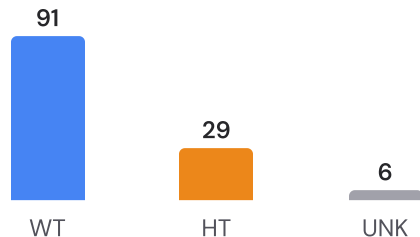
5
years

DATASET

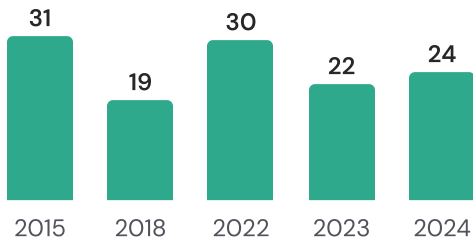
Dataset composition

Raw corpus · 126 mice · 35 dams · 125,576 syllables

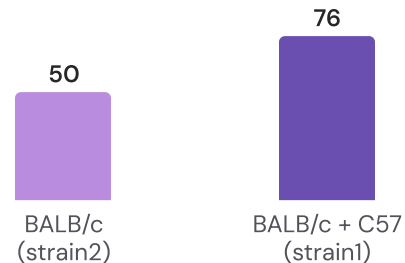
Mice by offspring genotype



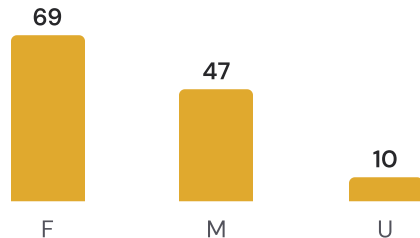
Mice by year



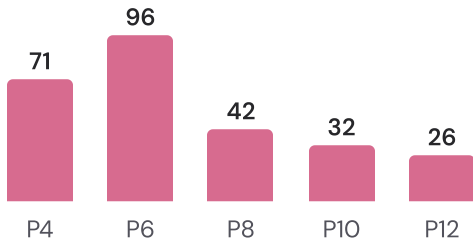
Mice by strain



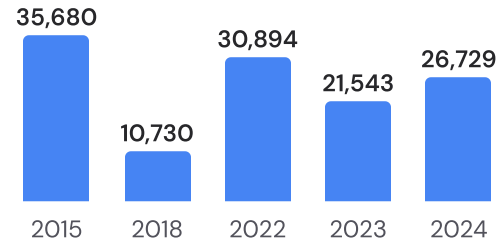
Mice by sex



Mice by postnatal day



Syllables by year

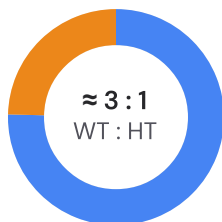


DATASET

Class balance & longitudinal structure

Recording-level class balance

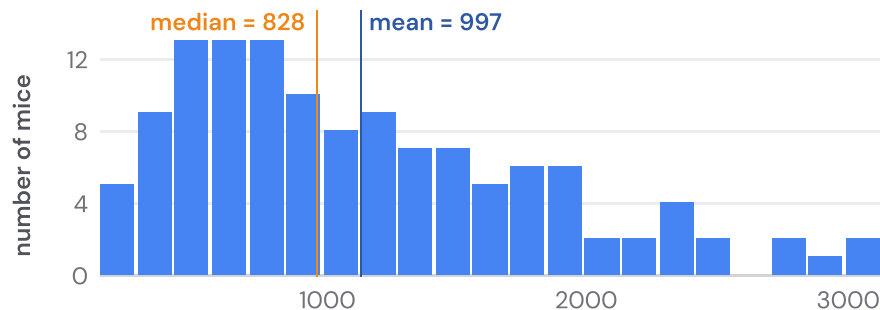
Baseline · 12,323 recordings



■ WT · 9,283 75.3%

■ HT · 3,040 24.7%

Syllables per mouse (n = 126)



syllables per mouse

One mouse contributes hundreds of correlated syllables.

DATASET

Strain: a biological confound

● BALB/c · pure

50 mice

Recorded in 2015 and 2018 · labeled `strain2`

● BALB/c + C57 · mixed

76 mice

Recorded 2022–2024 · real cross on a C57BL/6 background · labeled `strain1`

Strain differs in **both genetic background and recording era** — so it is confounded with genotype. We treat it as an explicit cohort scope and handle it carefully in the discussion.

SECTION 06

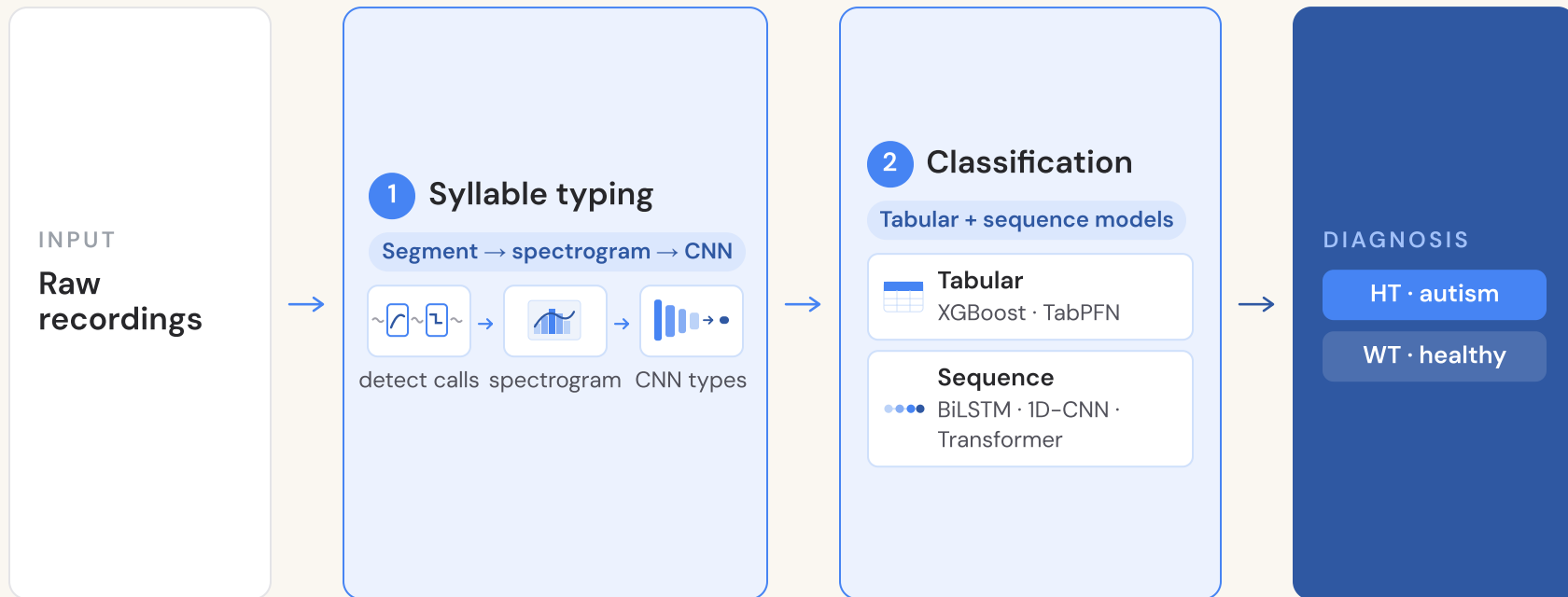
Methods

From raw ultrasonic audio to model-ready spectrograms.

06

From recordings to a diagnosis

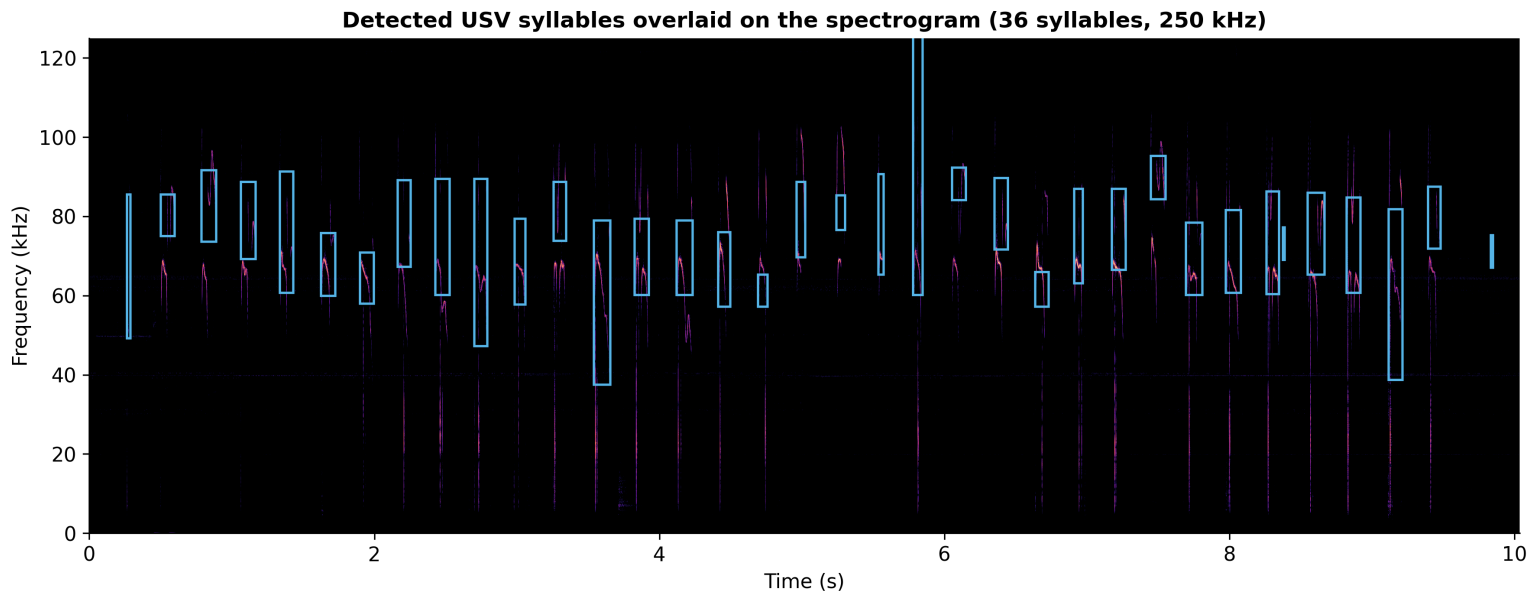
Every recording passes through two learned stages before a prediction.



Stage 1 turns audio into typed syllables · Stage 2 turns a subject's syllables into a WT-vs-HT prediction.

Finding the calls

A classical energy detector — not a network — locates every ultrasonic call in each recording.



Detected calls (blue boxes) overlaid on a 10-second recording — 36 syllables located automatically.

90 gammatone filters · 35–125 kHz

Tonal vs noise per 6 ms frame

Merge gaps < 20 ms · drop calls < 10 ms

Output: start · end · duration

Typing each call with a CNN

Syllable typing is a computer-vision task — each call becomes a spectrogram image the network classifies.

Syllable clip

One detected call, padded to 0.25 s and centered so only its shape matters.



Spectrogram image

128×128×3

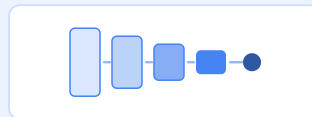
High-pass 30 kHz · STFT
win 512 / hop 128 · resize ·
normalize to 3 channels.



CNN backbone

BiT ResNet-50×3

~211M params ·
GroupNorm + Weight
Standardization ·
pretrained on ImageNet-
21k.



Syllable type

Softmax over 10 types ·
confidence < 0.5 →
Undefined.



A strong ImageNet backbone, fine-tuned on hand-labeled USV spectrograms — so it learns the call shapes from limited labeled data.

Model card: the syllable-typing CNN

What the network is, how it was trained, and what it outputs.

MODEL CARD

BiT ResNet-50×3

Big Transfer · ResNet-v2 bottleneck · width ×3

~211M

trainable parameters

128×128×3

input spectrogram (RGB)

TASK

Multi-class syllable typing — a spectrogram image → one call type

BACKBONE

ResNet-v2 bottleneck · blocks (3, 4, 6, 3) · width ×3

NORMALIZATION

Group Norm + Weight Standardization — the BiT signature

PRETRAINING

ImageNet-21k, then fine-tuned with a new Dense(→10) head on USV spectrograms

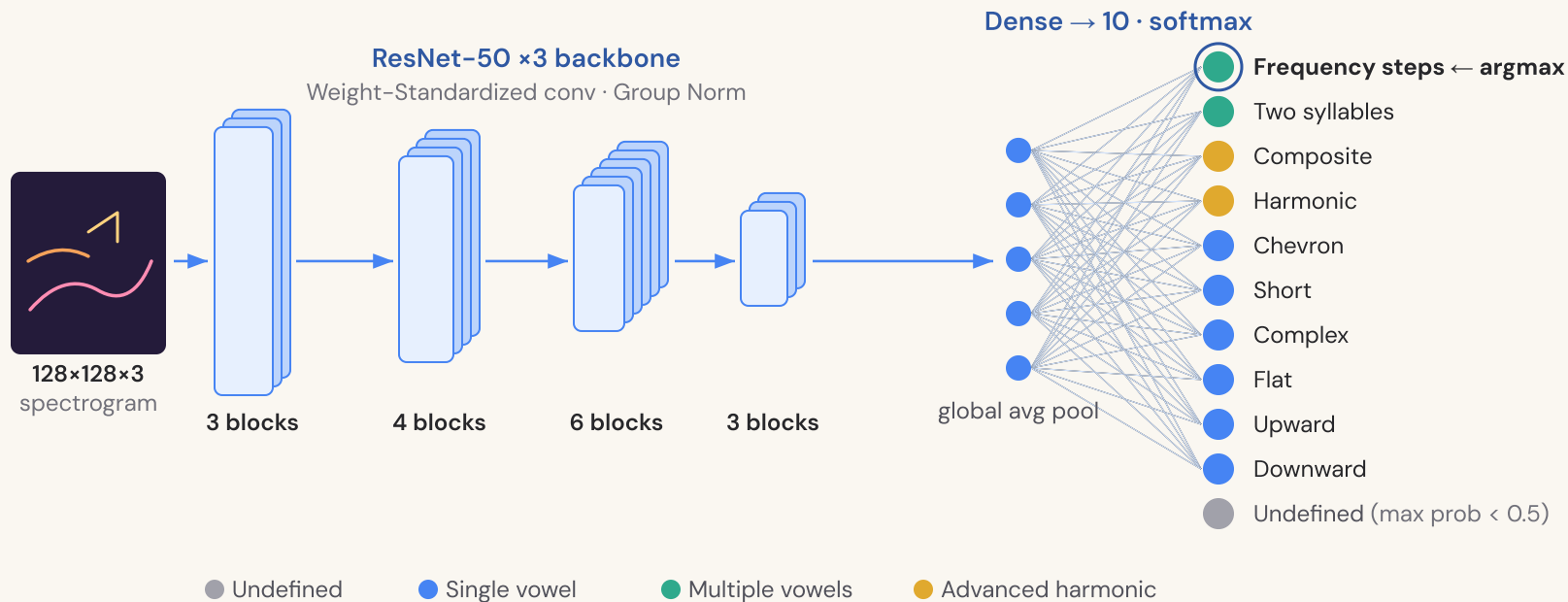
OUTPUT

Softmax over 10 types · confidence < 0.5 → **Undefined** (11th class, post-hoc)

INFERENCE

Batched (chunk 2048 · batch 32) — tens of thousands of syllables in minutes

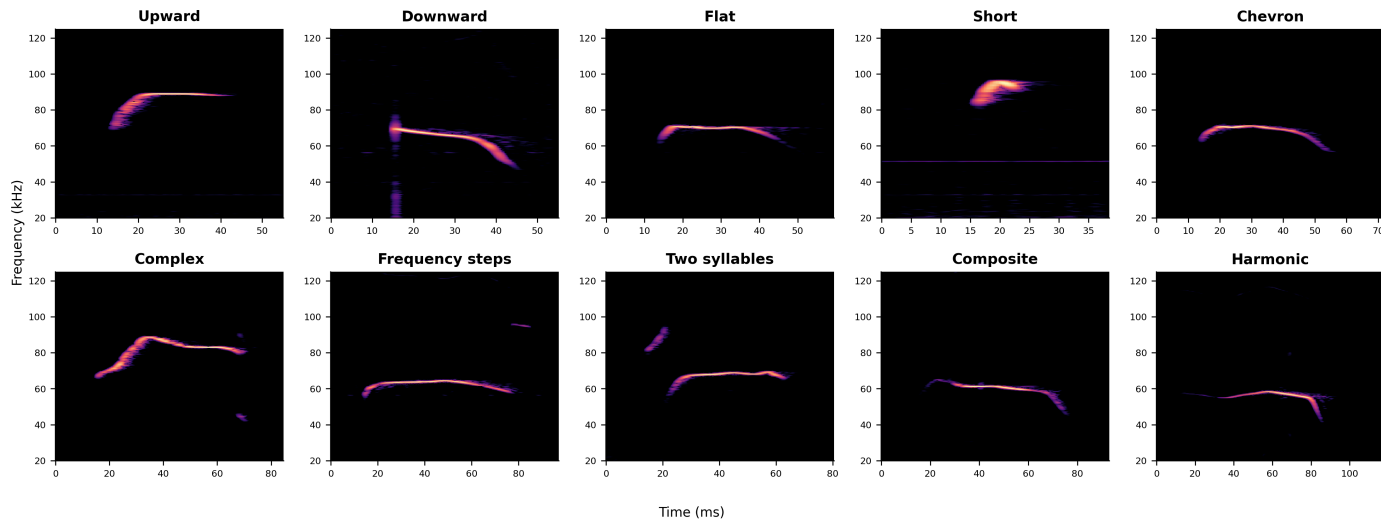
The network, layer by layer



A repertoire of 10 call types

Each call is one of ten shapes — which the CNN reads from its spectrogram — grouped into four complexity levels.

USV syllable-type taxonomy — representative real spectrograms (250 kHz)



■ **Single vowel**

Upward · Downward · Flat · Short
· Chevron · Complex

■ **Multiple vowels**

Frequency steps · Two syllables

■ **Advanced harmonic**

Composite · Harmonic

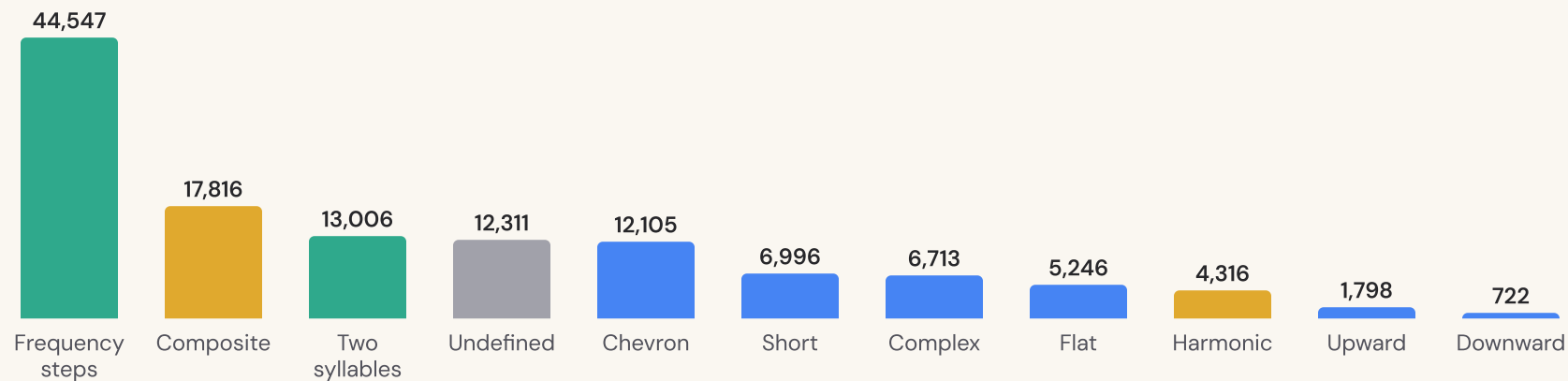
■ **Undefined**

Low-confidence calls

Syllable-type distribution

125,576 syllables, typed by the CNN and colored by complexity group.

■ Single vowel ■ Multiple vowels ■ Advanced harmonic ■ Undefined

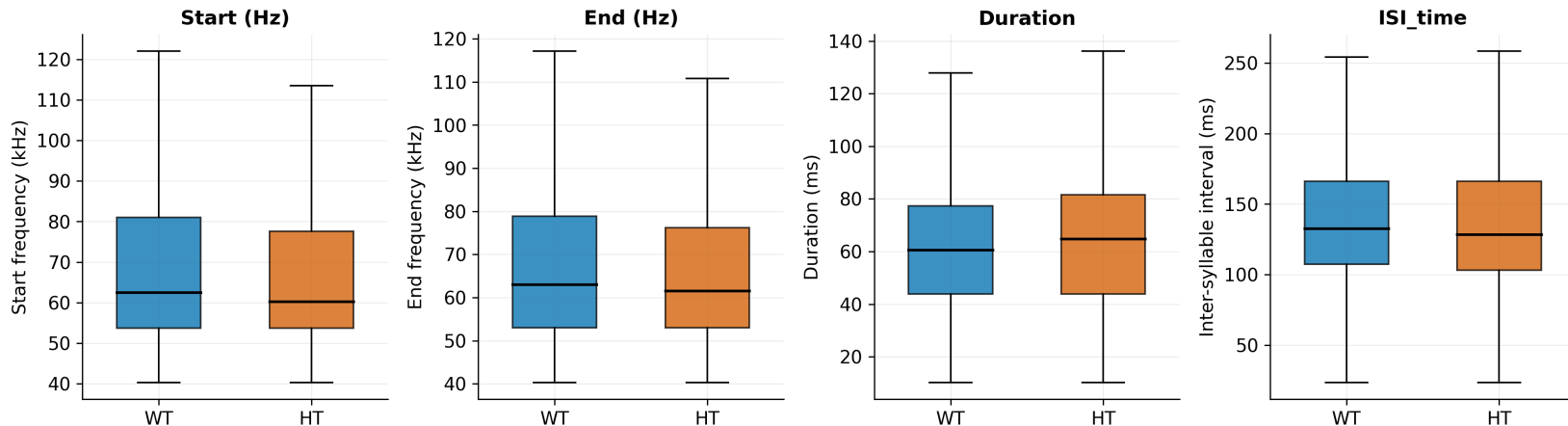


'Frequency steps' alone accounts for 44,547 calls. 'Undefined' (~10%) is dropped in the tabular path but kept in the sequence path.

The acoustic signal: WT vs HT

Per-syllable features differ subtly but consistently by genotype — the ASD-sensitive cues from Shekel et al. (2021).

Acoustic-feature distributions by genotype (syllable level)



SECTION 07

Models

Turning the syllable fingerprint into a WT-vs-HT prediction —
tabular vs sequence classifiers.



Two families of classifiers

The typed syllables feed two representations — and two very different model families.

Tabular models

Our focus

Recording matrix · $12,323 \times 48$

XGBoost · inherited baseline

XGBoost-tuned · new

TabPFN-3 · new

Strong results — the main story of this presentation.

Sequence models

Explored

Per-session sequences · $408 \times \leq 256$

BiLSTM

1D-CNN

Transformer

Did not outperform the tabular path — summarized, not deepened.

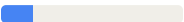
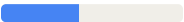
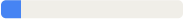
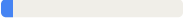
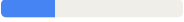
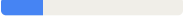
One shared evaluation protocol · subject-dependent vs subject-independent · per-strain vs pooled

One recording → 48 numbers

Each recording is reduced to a fixed 48-column vector — which averages away the temporal order of the calls.

10 syllable types × 4 features = 40

Example · one recording

Syllable type	Mean start kHz	Mean end kHz	Relative freq	Mean dur. ms
 Complex	68	61	 6%	78
 Frequency steps	63	60	 35%	62
 Composite	62	58	 15%	84
 Two syllables	66	63	 11%	66
 Upward	74	84	 4%	41
 Flat	60	59	 5%	33
 Harmonic	58	54	 4%	96
 Downward	78	66	 2%	45
 Chevron	66	70	 10%	52
 Short	79	77	 8%	18

METADATA · 6

Avg ISI

Sex

Age

Session

Strain

Mother genotype

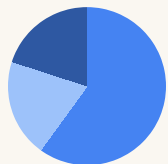
TARGET Pup genotype

Mouse index grouping only

EVALUATION

Two ways to split the data

Same 106 mice — what changes is **what you split**. ● WT · 82 ● HT · 24



- Train 60%
- Val 20%
- Test 20%

same 60/20/20
in both

Subject-dependent

Split by **recording**.

106 mice



Every mouse feeds all three splits — so the **same animal appears in Train and Test**.
Optimistic.

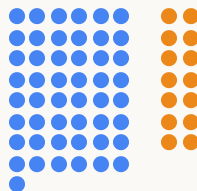
within-subject: classify a new recording of a *known* animal.

Subject-independent

Split by **mouse** · stratified · no overlap.

Train · 63

49 WT · 14 HT



Val · 21

16 WT · 5 HT



Test · 22

17 WT · 5 HT



Only 24 HT mice in total → ≈ 5 land in Test → **high-variance folds**.

across-subject: classify a *completely unseen* animal.

Model card: XGBoost baseline

MODEL CARD · INHERITED

XGBoost

Gradient-boosted trees · the baseline we inherited

50

trees · depth 5 · lr 0.1

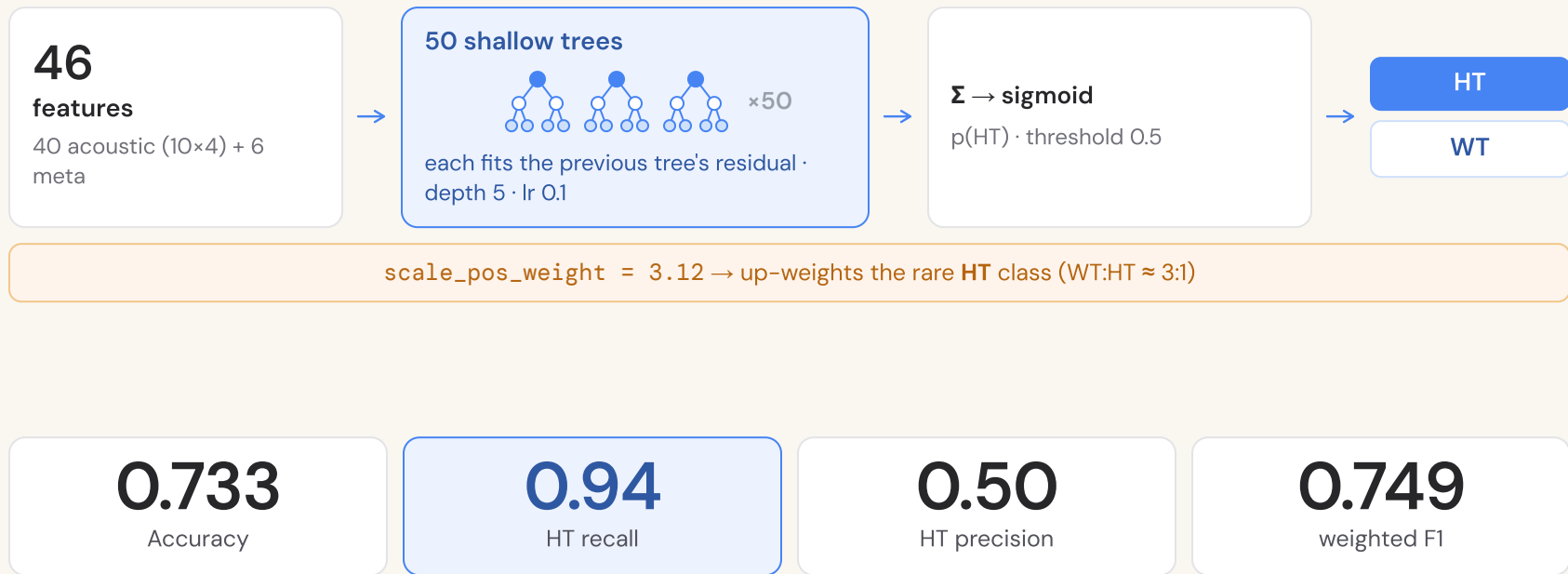
1

the only model at handover

TYPE	Gradient-boosted decision trees (ensemble)
RECIPE	The old, un-tuned config shipped with the project
TREES	50 · max_depth 5 · learning rate 0.1
IMBALANCE	scale_pos_weight = $n_{WT} / n_{HT} \approx 3.1$ (manual)
ROLE	The single model that existed at handover — our reference point
STATUS	The starting point for everything that follows

How gradient boosting decides

The inherited baseline — with its real test numbers.

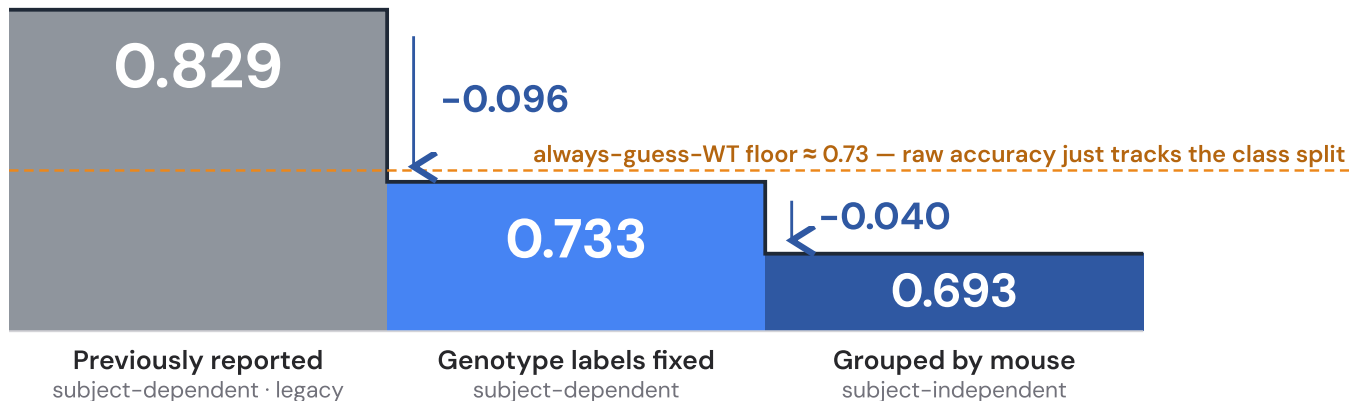


inherited XGBoost · subject-dependent eval · test WT 74% / HT 26%

RESULTS

Baseline

The inherited XGBoost, re-measured on corrected labels — the same model under **two different evaluation regimes**.

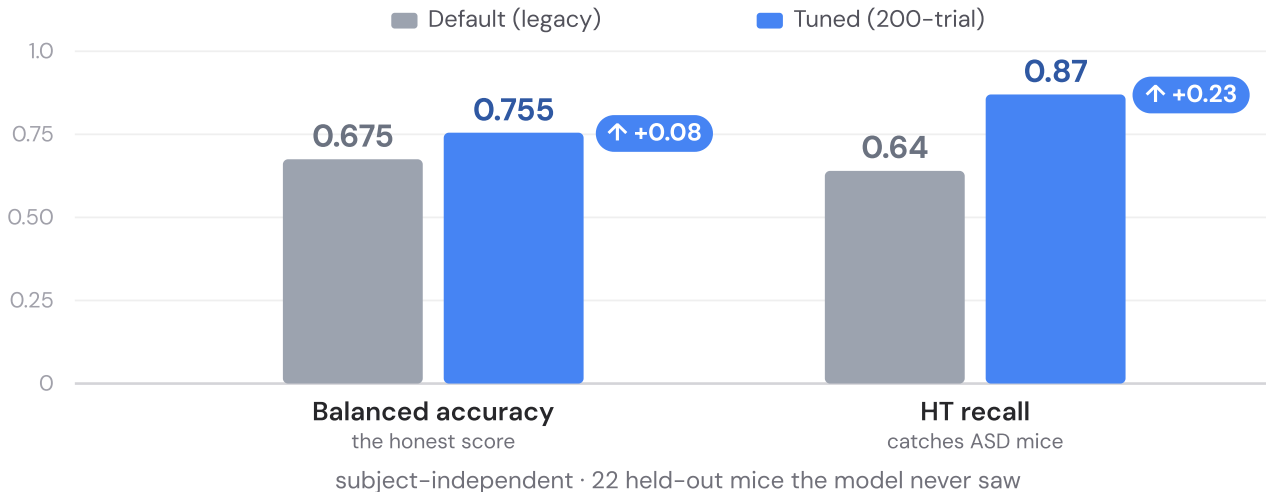


TWO REGIMES, TWO QUESTIONS

Subject-dependent and subject-independent test **seen vs unseen animals** — two different questions, not better-and-worse.

Why tuned beats default

In the regime that counts — **unseen mice** — tuning kills the overfit.



LEANER TREES

50 · depth 5 → **20 · depth 3**

STRONGER REGULARIZATION

min_child_weight **20** · reg_lambda **3.0**

FOUND BY SEARCH

200-trial random · cross-validated

TAKEAWAY

Same features, same mice — **tuning alone turned memorization into generalization.**

Three tabular classifiers

From the inherited baseline to a tabular foundation model.

Inherited

XGBoost baseline

Gradient-boosted trees on the old, un-tuned recipe shipped with the project. Imbalance handled with $\text{scale_pos_weight} \approx 3.1$. The only model at handover — our reference point.



New

XGBoost tuned

A 200-trial randomized hyper-parameter search with cross-validation — selecting a distinct recipe for each split regime.



New

TabPFN-3

A prior-data-fitted transformer: in-context Bayesian inference in a single forward pass, with no hyper-parameter tuning. Our strongest tabular model.



What is TabPFN?

A **foundation model for tables** — it learns to predict during pre-training, then classifies your data in a single forward pass.

CLASSIC ML · E.G. XGBOOST



Your table



Train a new model — minutes to hours



Search hyper-parameters



Predict

A fresh model, fitted from scratch for every dataset.

vs

TABPFN · TABULAR FOUNDATION MODEL



Pre-trained **once** on millions of synthetic tables



Feed labeled rows + query as **context** — no fitting



One forward pass · no tuning



Predict **WT / HT**

Brings the learning with it — in-context, like an LLM for tables.

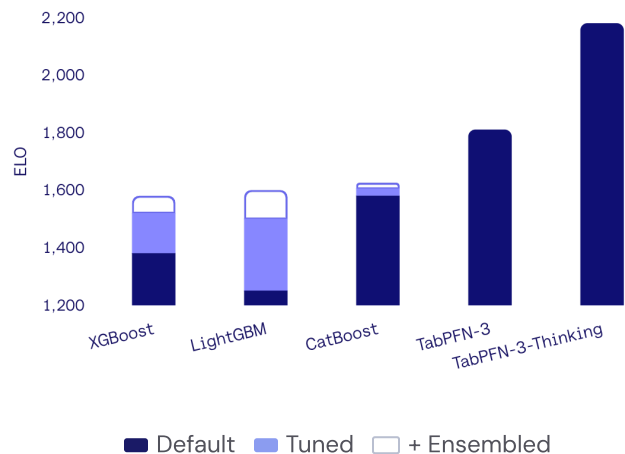
IN ONE LINE

A **language model for spreadsheets** — pre-trained to predict, it reads your labeled rows as a prompt and answers in one shot. No training, no tuning — which is exactly why it shines on small data like ours.

What is TabPFN?

A **tabular foundation model** — pre-trained once, it then classifies your data in a single forward pass.

Average accuracy across 51 OpenML datasets



TabPFN-3-Thinking leads the best boosted trees by ~420 ELO.



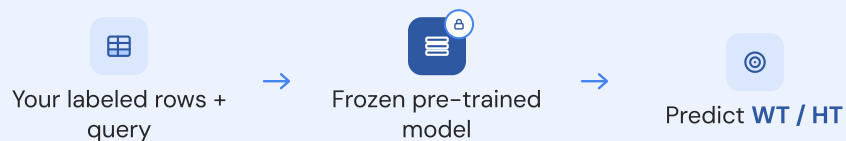
THE WAY IT WORKS TODAY

a new model every dataset



THE TABPFN WAY

pre-trained once · no tuning



93% win rate over classic ML

on TabArena — the leading structured-data benchmark

0.2s to predict 1M samples

fast inference for real-world workflows

80% of cases beat AutoML

TabPFN-3-Plus · Thinking mode (+420 ELO)

Model card: TabPFN-3

MODEL CARD

TabPFN-3

Prior-data-fitted transformer for tabular data

1

forward pass — no training loop

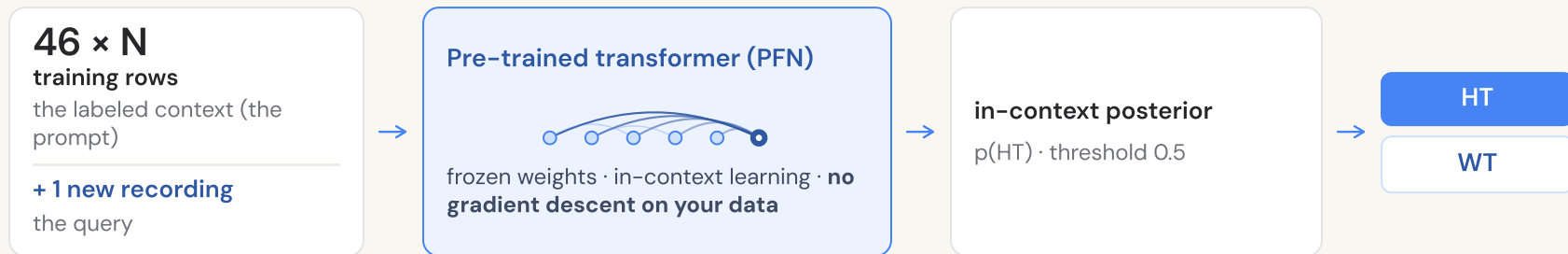
0

hyper-parameters to tune

TYPE	Foundation model for tabular data (prior-data-fitted network)
INFERENCE	In-context Bayesian prediction in a single forward pass
TUNING	None — no hyper-parameter search required
DATA USAGE	Folds its validation split into training — sees $\approx 80\%$ of data vs XGBoost's $\approx 60\%$
INPUT	The 48-column recording matrix (10 types $\times$ 4 features + 8 meta)
ROLE	Our best-performing tabular classifier

How TabPFN decides — without training

A transformer pre-trained once on millions of synthetic tables — no fitting, one forward pass.



No fitting on your data · no hyper-parameter tuning · `balance_probabilities = True` → native 3:1 class-imbalance correction

0.781

Accuracy

+0.048

0.92

HT recall

-0.02

0.55

HT precision

+0.05

0.794

weighted F1

+0.045

Best model in **both regimes** — barely drops on unseen mice: **0.729** subject-independent, vs XGBoost's 0.693.

What if the **order** of the calls is the signal?

The tabular models average each recording, discarding order and rhythm. So we looked at a session as a sequence.

ONE SESSION ≈ 10 MINUTES OF A PUP'S ISOLATION CALLS, IN THE ORDER THEY HAPPEN



each call →  14 numbers — 4 acoustic · 8 type embedding · 2 flags · up to 256 steps per session

TABULAR TRACK · ORDER DISCARDED



Collapses to **per-type averages** — the sequence is gone.

SEQUENCE TRACK · ORDER PRESERVED



Keep the calls **in order** — the melody, not just the notes.

Gal et al. (2023): the temporal dynamics of these calls carry phenotypic signal.

Models tried:

BiLSTM

1D-CNN

Transformer

A lucky fold, not a real gain

One split looked excellent; cross-validation showed the cause was data scale — not the hypothesis.

BALANCED ACCURACY · UNSEEN MICE

0.704

best single split
looked fantastic

-0.141



5-fold
grouped CV

0.563

± 0.063

honest estimate

Chance = 0.50 · some models even predicted HT for **every** session — a score that looks high but has no real skill.

WHY · DATA STARVATION

125,576 syllables



408 sessions



~19 HT / fold

Only 24 HT mice in the whole dataset

Honest conclusion — **not supported at this data scale**, not that temporal order is uninformative. With more animals, the dynamics reported by Gal et al. may well be learnable.

VERDICT

Explored, not chosen

SECTION 08

Results

Performance, discrimination, and comparison to baselines.

08

Everything we evaluated

Every model, on each representation, across cohort scope and split regime — plus the search, sweep and validation behind them.

	POOLED		STRAIN 1		STRAIN 2	
	dep	ind	dep	ind	dep	ind
TABULAR MODELS 48-feature per-recording aggregates						
XGBoost inherited baseline	●	●	●	●	●	●
XGBoost-tuned 200-trial search	●	●	●	●	●	●
TabPFN-3 foundation model	●	●	●	●	●	●
SEQUENCE MODELS 256×14 ordered-syllable tensors						
BiLSTM 2-layer bidirectional	●	●	.	.	.	.
1D-CNN conv + masked pool	●	●	.	.	.	.
Transformer CLS + 2 encoder layers	●	●	.	.	.	.

24

reported model × cohort × split evaluations

AND, BEHIND THEM

200

trial search
hyperparameters · per regime

8

imbalance levers
× 3 sequence architectures

5

fold grouped CV
mice separate across folds

1

CNN syllable typer
BiT ResNet-50×3 · transfer learning

RESULTS

Two numbers, honestly reported

TabPFN reaches the **highest accuracy** of our tabular models — and it beats the inherited baseline in both regimes.

SUBJECT-DEPENDENT

0.781

accuracy on animals seen before · up to 0.801 per cohort

SUBJECT-INDEPENDENT

0.729

accuracy on completely unseen animals · weighted F1
0.743 · ROC-AUC 0.783

Inherited XGBoost baseline, for reference: **0.733** subject-dependent · **0.693** subject-independent

RESULTS

The tabular models, side by side

Recording level, grouped by split regime: **seen** vs **unseen** animals. TabPFN leads overall; tuned-XGBoost gives the best minority recall on unseen animals.

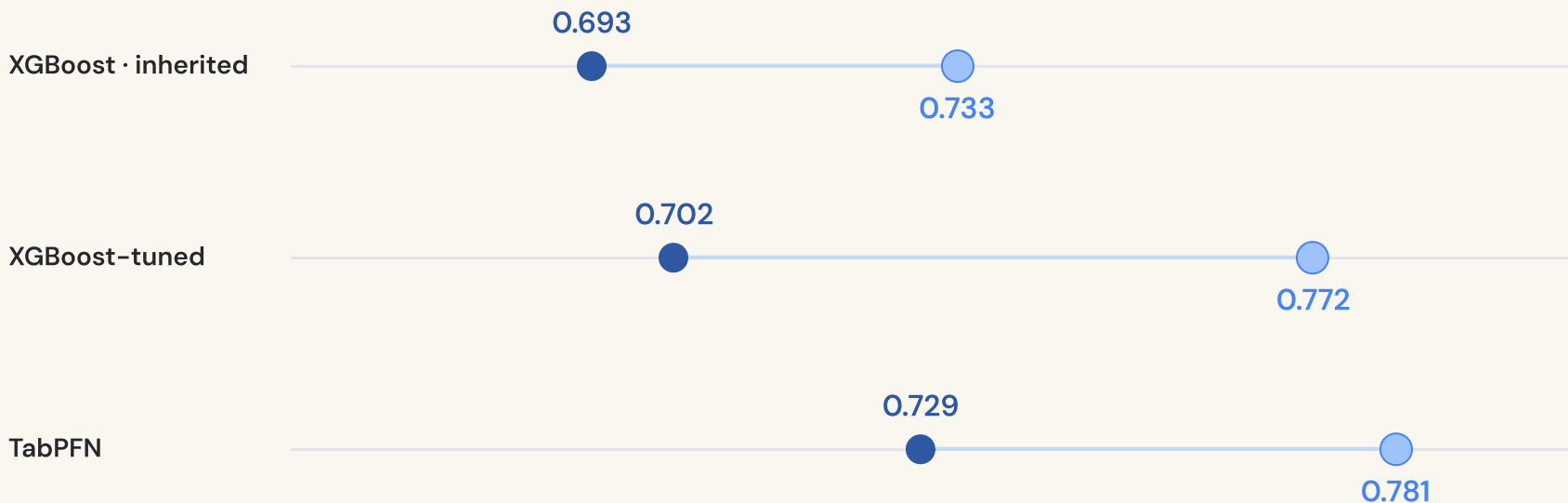
Model	Acc	Wtd F1	Bal. acc	ROC-AUC	HT recall	HT prec	HT F1
● Subject-dependent · seen animals — same mouse in train & test							
XGBoost · inherited	0.733	0.749	0.795	0.876	0.940	0.496	0.649
XGBoost-tuned	0.772	0.785	0.798	0.885	0.844	0.543	0.661
TabPFN	0.781	0.794	0.828	0.908	0.918	0.550	0.688
● Subject-independent · completely unseen animals							
XGBoost · inherited	0.693	0.706	0.678	0.770	0.637	0.452	0.529
XGBoost-tuned	0.702	0.719	0.725	0.753	0.869	0.473	0.612
TabPFN	0.729	0.743	0.662	0.783	0.782	0.499	0.610

☐ TabPFN — best overall ☐ best HT recall in each regime Positive class = HT (Mthfr model-ASD)

RESULTS

The generalization gap

Every model drops from seen animals to unseen — but TabPFN transfers best.

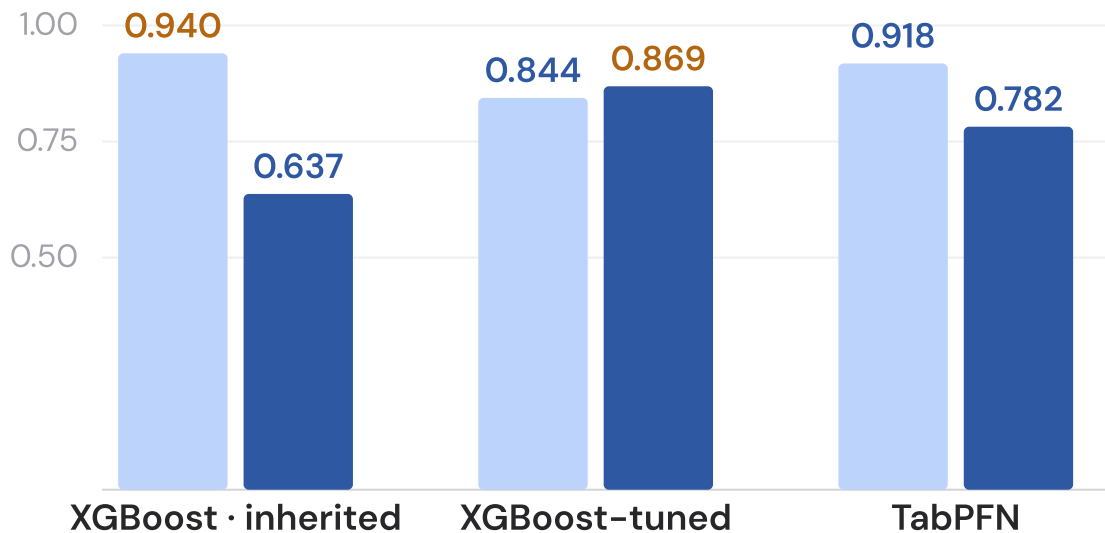


● subject-dependent · seen animals ● subject-independent · unseen animals accuracy scale 0.66 – 0.80

RESULTS

Catching the true cases: HT recall

Recall = the share of real model-ASD pups a model flags. All models score high on seen animals — but on **unseen** ones the ranking flips.



■ dependent · seen ■ independent · unseen ■ best recall in regime

On unseen animals the inherited model collapses to **0.637** — tuning restores recall to **0.869**, the best of any model.

RESULTS

The clinically important number

A screening tool must not miss true cases — so HT recall on unseen animals matters most.

HT RECALL · UNSEEN ANIMALS

0.869

tuned XGBoost catches ~87% of true model-ASD pups ·
TabPFN holds 0.782

BEST MODEL DEPENDS ON THE GOAL

Overall accuracy

→ TabPFN

Minority-class recall (screening)

→ tuned XGBoost

Where each true class goes

Each bar splits a true class into **correctly classified** vs **confused**. True HT is mostly caught; true WT always leaks ~30% into HT — the precision wall.

XGBoost · inherited

misses 36% of cases

true WT · n = 1,560

71%

correct WT (specificity)

29%

missed

true HT · n = 579

64%

correct HT (recall)

36%

missed

XGBoost-tuned

best recall · misses
13%

true WT · n = 1,560

64%

correct WT (specificity)

36%

missed

true HT · n = 579

87%

correct HT (recall)

13%

missed

TabPFN

misses 22% of cases

true WT · n = 1,560

71%

correct WT (specificity)

29%

missed

true HT · n = 579

78%

correct HT (recall)

22%

missed

Tuning cuts **missed cases 36% → 13%** — but ~30% of healthy pups are still flagged HT: the precision wall, a feature-level limit.

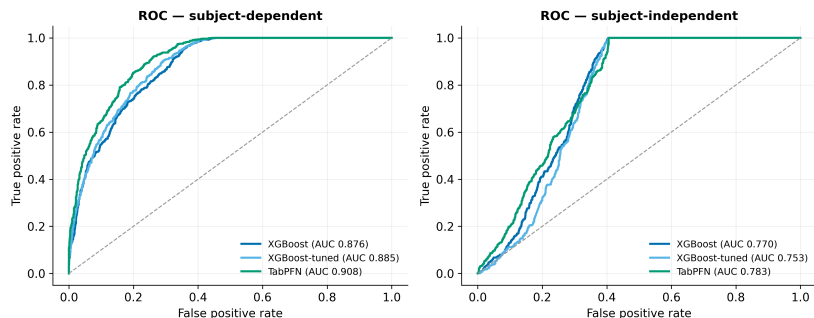
RESULTS

Ranking quality: ROC and precision–recall

Ranking is strong and well above chance — but the regime shift and the minority-class precision decay both show here.

ROC

ROC curves — headline tabular models

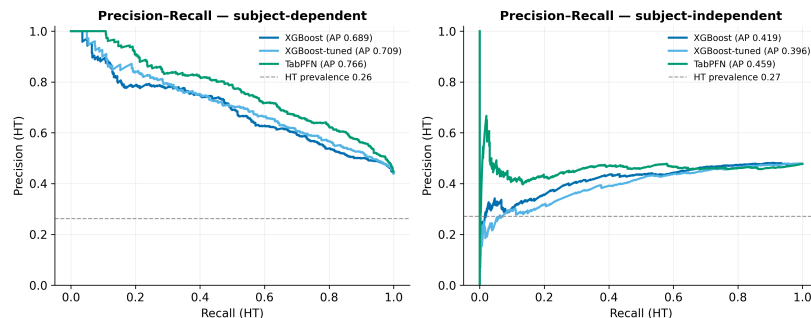


TabPFN ranks best

ROC-AUC 0.908 seen → 0.783 unseen

Precision–recall · HT

Precision-Recall curves (minority HT class)



Regime cost ≈ -0.12 AUC

the generalization gap, quantified

Above prevalence (0.26)

real signal — but precision decays at high recall

RESULTS

What the signal can and can't do

Two ceilings that shape every conclusion.

THE HT PRECISION WALL

~0.50

HT precision is pinned near 0.50 across every model, tuning and split — about half of positive alerts are false. A **feature-level ceiling**, not the classifier.

STRAIN = COHORT CONFOUND

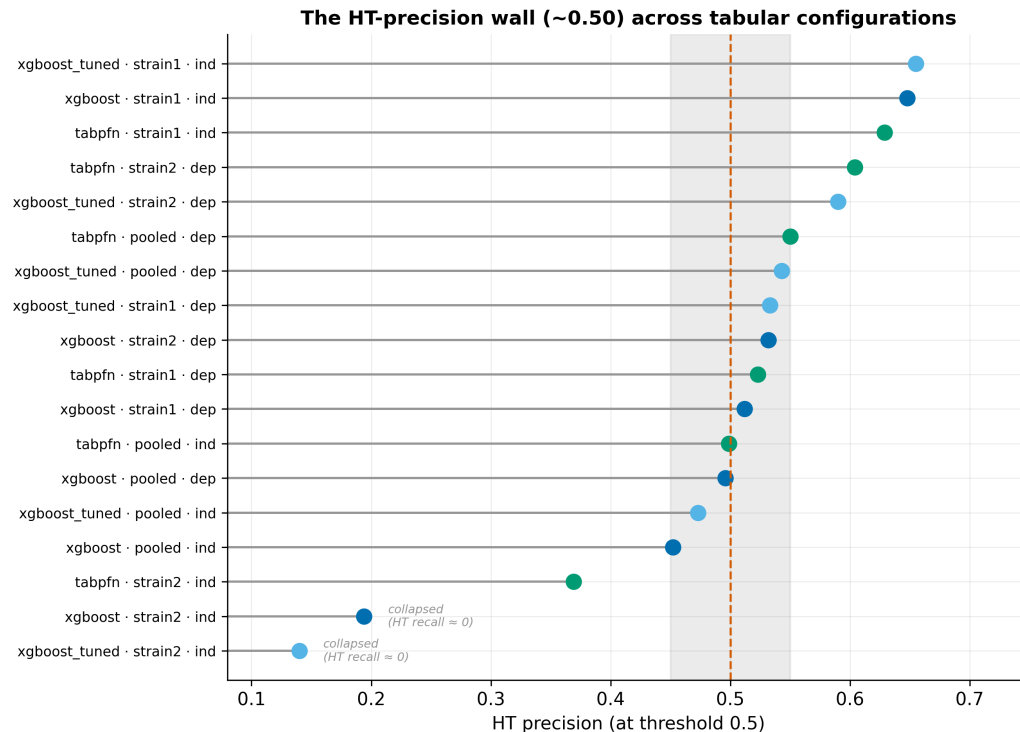
0.903 strain1 across-subject

...while strain2 collapses. Strain is confounded with both **recording era** and **genetic background** — so per-strain numbers can't be read as pure phenotype.

RESULTS

The HT precision wall

HT precision clusters around 0.50 across every tabular configuration — a feature-level ceiling, not a classifier limit.



RESULTS

Which features drive the call?

Averaged XGBoost gain importance — metadata dominates; among acoustic cues, border frequencies and per-type durations lead.



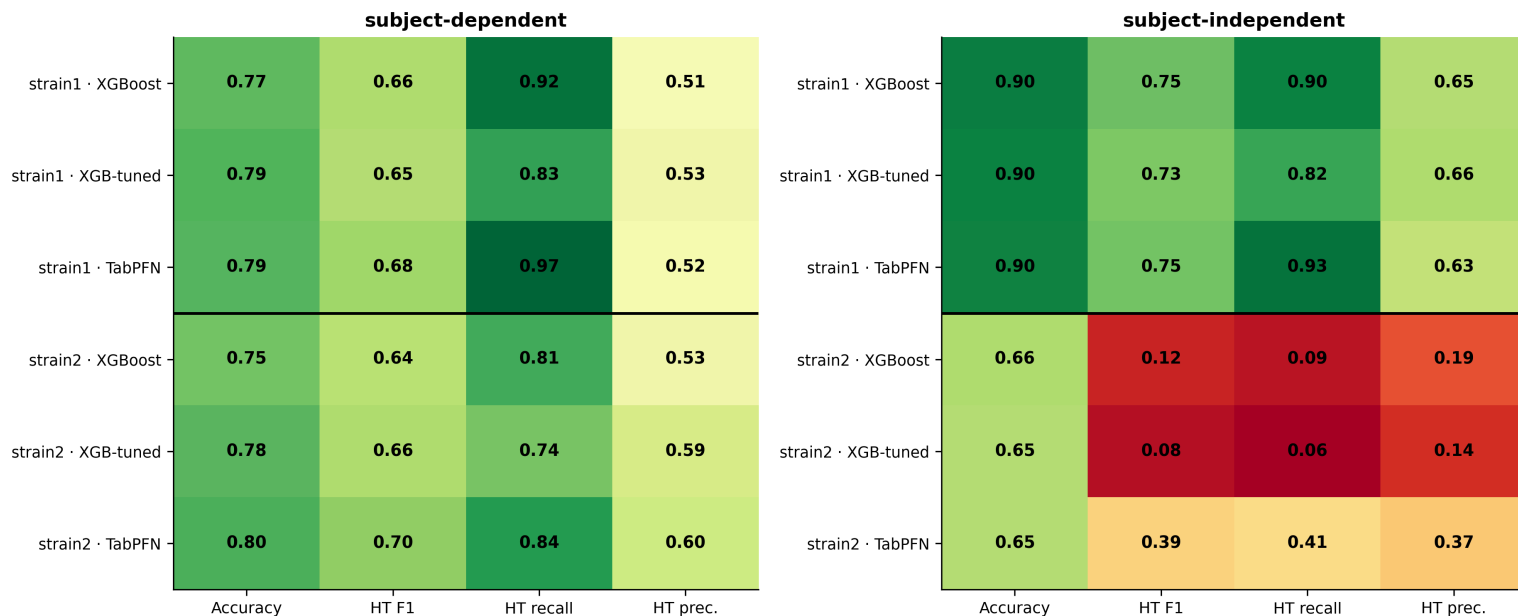
■ metadata ■ acoustic feature much of the signal is metadata — part of why the precision wall holds

RESULTS

Strain confounds the phenotype

strain1 (the recent, mixed-background cohort) is separable; strain2 collapses — strain is entangled with cohort and genetic background.

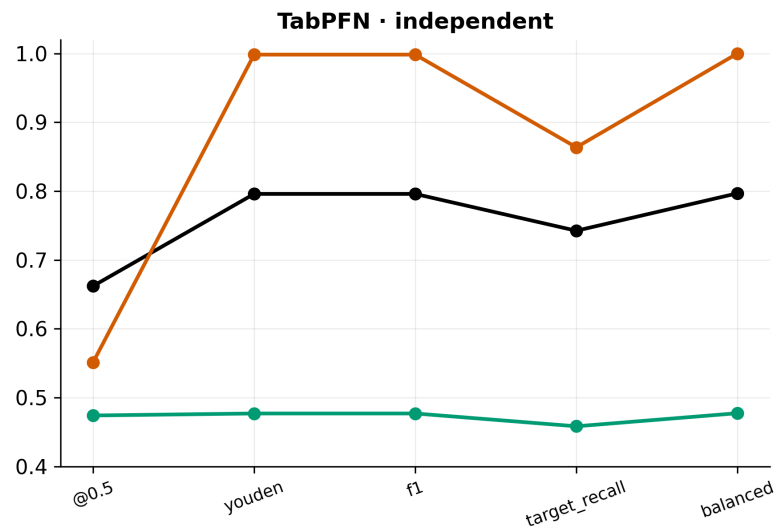
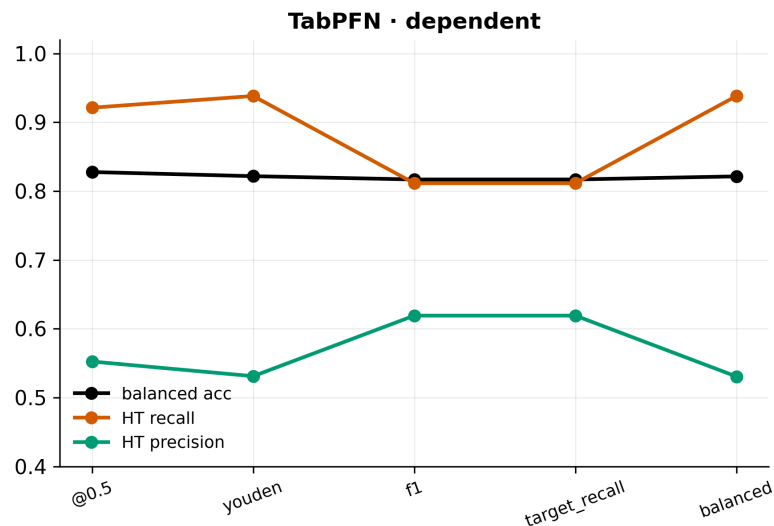
Per-strain results (all tabular models) — strong strain1 vs collapsing strain2



The threshold is a dial we control

The precision wall is fixed — but the decision threshold is ours. For a screening tool we can trade toward high recall.

Decision-threshold objectives — moving the operating point (TabPFN)



Same model, moved point

one TabPFN — @0.5, Youden, F1, target-recall, balanced

Recall → ~1.0 on unseen

push the operating point to catch nearly every case

Precision holds ~0.47

the wall stays — so choose recall for screening

SECTION 09

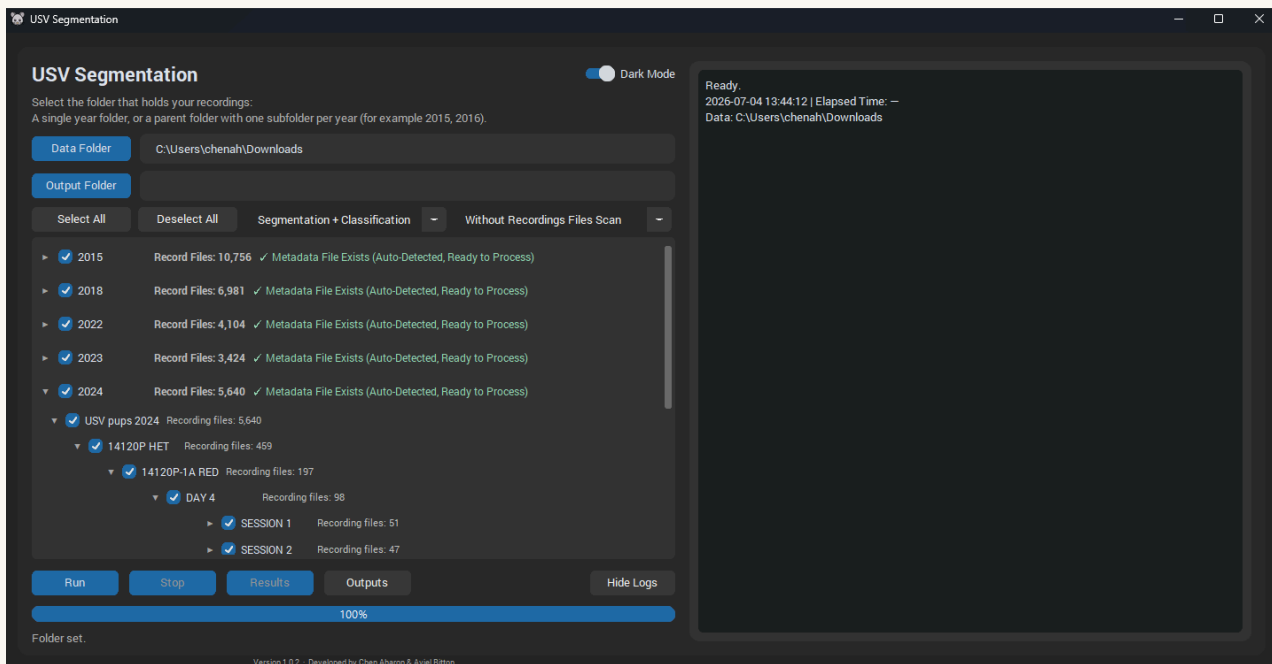
Engineering contribution

By HIT's guidelines, turning an inherited research prototype into a reproducible, end-to-end system is a result in its own right.



A segmentation app the lab actually uses

A one-click GUI wrapping our segmentation + CNN typing — built for Prof. Hava Golan to curate USVs without the command line.

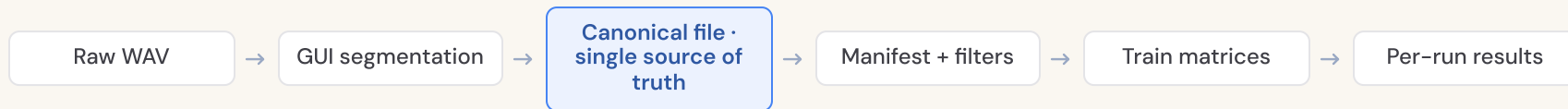


- **year** → **session** hierarchy
- Same algorithms as the **pipeline**
- Outputs the **canonical file**
- Caught the **genotype error**

Published & citable — **USV Segmentation v1.0.2** · Zenodo · one-click Windows installer

From inherited prototype to a single source of truth

One documented data lineage, versioned end to end — rebuilt on top of an inherited research prototype.



Single source of truth

One canonical file + a versioned manifest replace scattered, undocumented spreadsheets — every number traces back to a specific run.



Data-integrity fix

Establishing that source exposed a genotype-labeling error: **14 mice · 2,495 rows** corrected (HET → WT) — and with it, the honest, lower baseline.



End-to-end reproducible

Runs from a clean checkout: pinned environment (requirements + Dockerfile) and documented CLI flags for model, split regime and strain.

OPEN SOURCE Public GitHub repository — clone and run

CITABLE Zenodo · doi:10.5281/zenodo.20096810

SECTION 10

Conclusions & future work

Closing the research loop — what we answered, and what's next.



CONCLUSIONS

Answering the research questions

RQ1 MODELS

Yes

Do the new models beat the inherited baseline?

TabPFN wins both regimes (0.781 within · 0.729 across · ROC-AUC 0.783); tuned XGBoost gives the best minority recall (0.869 on unseen animals).

RQ2 FEATURES

Partly

Which acoustic features drive the classification?

Border frequencies and per-syllable-type durations (consistent with Shekel et al.) — but HT precision ≈ 0.50 marks a feature-representation ceiling.

RQ3 REGIME & GENERALIZATION

Decisive

How do split regime and grouping affect generalization?

Regime dominates — report within vs across separately (up to ~ 0.30 gap in HT recall); strain is confounded with cohort; sequence models don't help at this scale.

ENGINEERING

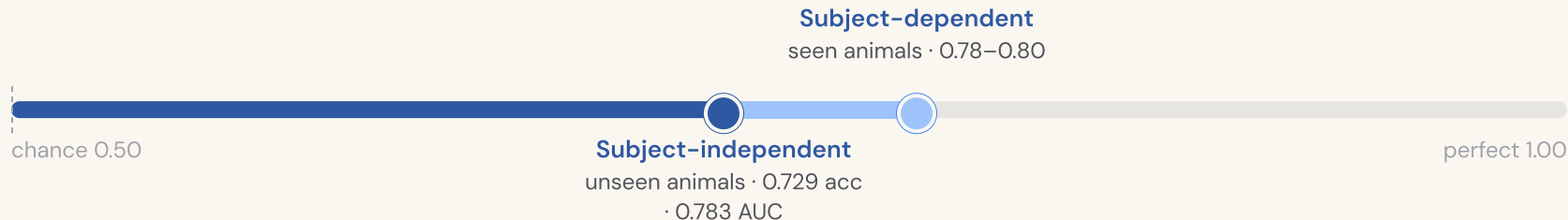
Yes

Can the inherited pipeline be made reproducible?

A reproducible, versioned, end-to-end pipeline with a single source of truth and GUI curation — which also exposed and fixed a genotype-labeling error.

CONCLUSION

Mouse USVs carry a **real** ASD-model signal — strong on known animals, **realistically** above chance on new ones.



The signal is real

USVs separate wild-type from the Mthfr ASD model — 0.869 HT recall catches true cases on unseen animals.



Honestly bounded

A ~0.50 feature-level precision ceiling and a strain-cohort confound — reported per regime, never hidden.



Reproducible & auditable

A single source of truth, open-sourced end to end — every number traceable, a foundation to build on.

Where this goes next

The three highest-leverage next steps.

1

Richer per-call features

Add full frequency-contour descriptors (mean / min / max / slope, bandwidth, FM depth) to test whether the HT-precision wall finally breaks — the highest-leverage experiment.

2

A true cross-cohort test

Train on one strain and evaluate on the other, separating the ASD phenotype from cohort membership and controlling the confirmed genetic-background difference.

3

Per-subject decisions

Aggregate call- and session-level predictions into one vote per mouse and report subject-level metrics — the clinically meaningful resolution.

THANK YOU

Questions & discussion

Advisor

Dr. Dror Lederman

Biology lead & data owner

Prof. Hava M. Golan

Ben-Gurion University

Presented by

Chen Aharon · Aviel Bitton

M.Sc. Data Science